

Data Analyst - Assignment

Objective

We are looking for an analyst who can do more than write SQL, build dashboards, or train models.

Your job is to determine whether a business claim is actually true when the underlying data is messy, incomplete, contradictory, and potentially misleading.

You will be given approximately **12 months of collections data** across borrowers, accounts, agents, campaigns, calls, digital interactions, field visits, promises-to-pay, and payments.

The business currently reports: **“Recovery has improved by 11% month-on-month.”**

The leadership team is not convinced.

Your assignment is simple:

Find out what actually happened.

You should assume that some of the data, definitions, attribution logic, and historical reporting may be wrong.

Business Context

The platform manages collections across:

- Human calling
- Predictive/progressive dialing
- IVR
- Agentic voice
- WhatsApp
- SMS
- Field visits

Leadership wants to understand whether operational performance is genuinely improving and where the next **₹10 Cr investment** should be deployed.

You have been given raw data from multiple systems.

There is **no guaranteed data dictionary**.

There is **no single source of truth**.

You cannot assume that the existing reporting logic is correct.

Your Mission

Answer four questions:

1. What happened?

Reconstruct the actual business performance over the last 12 months.

Determine:

- When performance changed
- What changed
- Where it changed
- Which metrics are genuinely improving
- Which metrics are misleading

2. Why did it happen?

Identify the major drivers of the change.

Investigate at minimum:

- Portfolio mix
- DPD
- Client
- Geography
- Language
- Agent
- Agent tenure
- Campaign
- Channel
- Telephony vendor
- Calling time
- Attempt frequency
- Borrower segment

Do not assume that correlation means causation.

Clearly classify conclusions as:

Fact / Strong Evidence / Correlation / Hypothesis

3. Is the reported 11% improvement real?

Build an independent definition of recovery performance.

You should challenge the existing definitions of:

- Contact rate
- RPC
- PTP rate
- PTP kept rate
- Recovery rate
- Recovery per account
- Recovery per agent-hour
- Cost per ₹ recovered
- Channel conversion

Explain why your definitions are appropriate.

If the reported 11% improvement is wrong, quantify the actual change.

If it is correct, prove it.

4. Where should we invest ₹10 Cr?

Leadership can invest the ₹10 Cr in only one area:

1. Better telephony infrastructure
2. More collection agents
3. AI voice automation
4. Better borrower targeting
5. WhatsApp/digital engagement
6. Field operations

Recommend **one**.

Your recommendation must include:

- Expected incremental recovery
- Estimated cost
- Expected ROI
- Break-even point
- Key assumptions
- Downside scenario
- Confidence/range around your estimate

If the data is insufficient to make a reliable recommendation, say so and explain what additional experiment/data would be required.

Data

You will receive data covering approximately 12 months.

Expected datasets include:

- `borrowers`
- `accounts`
- `agents`
- `agent_sessions`
- `campaigns`
- `daily_targeting`
- `calls`
- `call_attempts`
- `call_dispositions`
- `whatsapp_events`
- `sms_events`
- `field_visits`
- `promises_to_pay`
- `payments`
- `vendor_telephony`
- `complaints`
- `account_status_history`

The datasets may contain:

- Duplicate records
- Missing values
- Conflicting timestamps
- Different time zones
- Inconsistent IDs
- Late-arriving events
- Changed schemas
- Changed disposition codes
- Duplicate payment events
- Historical records being overwritten
- Multiple identifiers for the same agent
- Inconsistent campaign definitions

Do not assume the cleanest-looking table is the most reliable.

Part 1 - Build the Golden Dataset

Create a trustworthy analytical layer from the raw data.

You must document:

- Source-of-truth decisions
- Entity resolution
- Deduplication logic
- Missing-data treatment
- Timestamp treatment
- Payment attribution
- Historical changes
- Exclusion rules
- Data-quality issues
- Assumptions

Show:

Raw Records → Rejected/Corrected → Golden Dataset

Quantify the impact of your cleaning decisions.

Part 2 - Data Forensics

You should actively search for inconsistencies.

At minimum, investigate whether the data contains:

A. Duplicate payments

Could retries or ingestion issues be inflating recovery?

B. Attribution errors

Are payments being incorrectly attributed to the latest campaign or interaction?

C. Timezone problems

Are calls being classified into the wrong hour/day?

D. Vendor mapping changes

Did telephony response/disposition codes change during the period?

E. Agent identity problems

Does the same agent appear under multiple identifiers?

F. Portfolio mix changes

Did the business acquire a fundamentally different portfolio?

G. Denominator manipulation

Are unsuccessful accounts disappearing from the population used to calculate conversion?

You are **not being told which of these problems actually exist**.

Find them if they exist.

Part 3 - Statistical Investigation

You must investigate whether observed improvements are caused by operational changes or by changes in the underlying population.

At minimum, investigate:

- Mix effects
- Cohort effects
- Selection bias
- Survivorship bias
- Simpson's paradox
- Attribution-window bias
- Time-series effects

You do not need to use sophisticated ML.

A simple, transparent statistical method is preferred over a complex model that cannot be explained.

Part 4 - Counterfactual

Assume that the business changed its targeting strategy midway through the year.

Leadership asks:

“What would recovery have looked like if we had not changed the targeting strategy?”

Design a methodology to estimate this.

You may use approaches such as:

- Matching
- Regression
- Difference-in-differences
- Propensity scoring
- Uplift modelling
- Controlled experimentation
- Bayesian approaches

Clearly state:

- Your treatment group
- Control group
- Assumptions
- Confounding factors
- Identification strategy
- Limitations

You will receive more credit for **correct reasoning than methodological complexity**.

Part 5 - Production Analytics Design

Assume your analysis will become a production system used by the leadership team every day.

Design:

Raw → Staging → Clean → Golden → Feature → Metrics → Dashboard

Explain:

- Data contracts
- Primary keys
- Metric definitions
- Data lineage
- Incremental processing
- Late-arriving data
- Backfills
- Data-quality checks
- Monitoring

- Anomaly detection

The design should be practical enough for an engineering team to implement.

Deliverables

1. SQL Repository

Production-quality SQL containing:

- Data cleaning
- Transformations
- Metric calculations
- Analytical queries

Queries should be reproducible.

2. Analysis Notebook

Use Python, R, SQL, or a combination.

The notebook should show your reasoning, not just the final charts.

3. Golden Dataset

Provide the final analytical dataset or reproducible pipeline used to create it.

4. Data Quality Report

Show:

- Major data issues
- Detection methodology
- Treatment
- Business impact

5. Executive Dashboard

Build **one screen**.

A CEO should be able to understand the situation in approximately **60 seconds**.

Do not optimize for visual complexity.

Optimize for decisions.

6. Executive Memo

Maximum **2 pages**.

Answer:

What happened?

Why did it happen?

How confident are we?

What should we do?

What is the expected financial impact?

7. Architecture Diagram

Show how your analytical system would work in production.

Time & Submission

Recommended time: 72 hours from receiving the dataset.

You are not expected to spend every hour working on the assignment.

We are evaluating:

Depth of thinking > volume of work

Submit:

- Git repository

- Notebook
- SQL
- Dashboard
- Golden dataset/pipeline
- Executive memo
- Architecture diagram